

12. (2 points)

Given the following data for a book of business:

| <u>Cumulative Paid Claims (\$000)</u> |               |               |               |               |               |               |                |
|---------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|
| Accident                              | 6             | 18            | 30            | 42            | 54            | 66            | Exposure       |
| <u>Year</u>                           | <u>Months</u> | <u>Months</u> | <u>Months</u> | <u>Months</u> | <u>Months</u> | <u>Months</u> | <u>(\$000)</u> |
| 2011                                  |               |               | 2,000         | 2,250         | 2,500         | 2,750         | 90,000         |
| 2012                                  |               | 1,750         | 2,250         | 2,500         | 2,650         |               | 100,000        |
| 2013                                  | 1,750         | 3,250         | 4,000         | 4,750         |               |               | 130,000        |
| 2014                                  | 2,000         | 3,750         | 4,500         |               |               |               | 200,000        |
| 2015                                  | 3,000         | 4,000         |               |               |               |               | 250,000        |
| 2016                                  | 2,000         |               |               |               |               |               | 320,000        |

- The most recent diagonal is as of March 31, 2016.

a. (1 point)

Briefly describe four issues in the data above that should be addressed when implementing the over-dispersed Poisson (ODP) bootstrap model.

b. (1 point)

For each data issue identified in part a. above, briefly describe an adjustment to the ODP bootstrap model to address the issue.

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# EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

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| <b>QUESTION 12</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |
| <b>TOTAL POINT VALUE: 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>LEARNING OBJECTIVE(S): A8</b> |
| <b>SAMPLE ANSWERS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |
| <b>Part a: 1 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| <i>Sample Responses include any 4 of the following:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |
| <ul style="list-style-type: none"> <li>Increasing exposures</li> <li>Last diagonal is not a complete year</li> <li>First evaluation period is not a complete year</li> <li>Missing earlier data in triangle, AY11 at 6 + 18 months and AY12 at 6 months</li> <li>High development at AY 2013 age 42 could be an outlier</li> </ul>                                                                                                                                                                                                                                   |                                  |
| <b>Part b: 1 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| <i>Sample Responses to the responses in part a. include:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                  |
| <ul style="list-style-type: none"> <li>Should divide claims by exposure and model loss ratios</li> <li>Annualize the losses, calculate LDFs and de-annualize results</li> <li>Calculate LDFs as normal. Need to estimate AY 2016 losses by dividing in half, to approximate 50% earned in that year</li> <li>Do not use those years when calculating volume weighted LDFs</li> <li>Exclude outlier from the calculation of the loss development factors and residual calculation, but re-sample the corresponding incremental when re-sampling triangles.</li> </ul> |                                  |
| <b>EXAMINER'S REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                  |
| Candidates were expected to identify data issues and related model adjustments when using the ODP bootstrap model for reserving.                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |
| <b>Part a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |
| Candidates were expected to list four problems related to the data provided in the problem in the context of an ODP bootstrap model.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |
| Common errors include:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| <ul style="list-style-type: none"> <li>Mistaking incomplete data for year 2011 for an outlier</li> <li>Not taking the fact the last diagonal is 9 months only into consideration when evaluating outliers (2015 was often identified as an outlier)</li> <li>Not taking exposure growth into consideration when evaluating outliers</li> <li>Mixing up CY and AY</li> <li>Stating that the last diagonal has 3 months of incremental loss data (it has 9 months in AY other than 2016)</li> <li>Providing problems that were not in this specific data.</li> </ul>   |                                  |
| <b>Part b</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |
| Candidates were expected to list 4 adjustments to the ODP bootstrap model to address the issue mentioned in part a. The adjustments listed needed to address the issue mentioned in a.                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| Common errors include:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| <ul style="list-style-type: none"> <li>Struggling in providing the correct adjustment for the partial first development period and the partial last calendar year.</li> <li>Partial first development period: Failure to mention that the adjustment was for the 2016 AY</li> </ul>                                                                                                                                                                                                                                                                                  |                                  |

## EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

- Partial first development period: Stating that the final results needed to be multiplied by 2
- Mixing up the adjustments for partial first development period and partial last calendar year
- Partial last calendar year : Removing the last diagonal as a solution
- Partial last calendar year: Failure to mention that the claims in the most recent diagonal needed to be re-adjusted to a partial year after the simulation process
- Providing adjustments for problems that were not present in the data.